Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.63)

Metric

$$ds^2 = -Y(t,r)^4 h(t)^2 \left(\frac{\partial Y}{\partial t}\right)^2 \mathrm{d}t \otimes \mathrm{d}t + \frac{1}{Y(t,r)^4} \mathrm{d}r \otimes \mathrm{d}r + Y(t,r)^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + Y(t,r)^2 \sin(\theta)^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{\partial}{\partial t} Y(t,r)^2}\, Y(t,r)^2 h(t)\right) dt + \left(\frac{\sqrt{2}}{2\,Y(t,r)^2}\right) dr$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{\partial}{\partial t} Y(t,r)^2}\, Y(t,r)^2 h(t)\right) dt + \left(-\frac{\sqrt{2}}{2\,Y(t,r)^2}\right) dr$$

$$m = \left(\frac{1}{2}\sqrt{2}Y(t,r)\right) d\theta + \left(\frac{1}{2} i\sqrt{2}Y(t,r)\left|\sin(\theta)\right|\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}Y(t,r)\right) d\theta + \left(-\frac{1}{2} i\sqrt{2}Y(t,r)\left|\sin(\theta)\right|\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,Y(t,r)^2 h(t) \frac{\partial}{\partial t} Y(t,r)}\right) dt + \left(\frac{1}{2}\sqrt{2}Y(t,r)^2\right) dr$$

$$n = \left(-\frac{\sqrt{2}}{2\,Y\left(t,r\right)^{2}\,h\left(t\right)\,\frac{\partial}{\partial t}Y\left(t,r\right)}\right)\,dt + \left(-\frac{1}{2}\,\sqrt{2}Y\left(t,r\right)^{2}\right)\,dr$$

$$m = \left(\frac{\sqrt{2}}{2\,Y\left(t,r\right)}\right)\,d\theta + \left(\frac{i\,\sqrt{2}}{2\,Y\left(t,r\right)\left|\sin\left(\theta\right)\right|}\right)\,d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,Y\left(t,r\right)}\right)\,d\theta + \left(-\frac{i\,\sqrt{2}}{2\,Y\left(t,r\right)\left|\sin\left(\theta\right)\right|}\right)\,d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,Y\left(t,r\right)^{2}\,h\left(t\right)\,\frac{\partial}{\partial t}Y\left(t,r\right)}\,\partial_t + \frac{1}{2}\,\sqrt{2}Y\left(t,r\right)^{2}\,\partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\,Y\left(t,r\right)^{2}\,h\left(t\right)\,\frac{\partial}{\partial t}Y\left(t,r\right)}\,\partial_t - \frac{1}{2}\,\sqrt{2}Y\left(t,r\right)^{2}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,Y\left(t,r\right)}\,\partial_\theta + \frac{i\,\sqrt{2}}{2\,Y\left(t,r\right)\left|\sin\left(\theta\right)\right|}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,Y\left(t,r\right)}\,\partial_\theta - \frac{i\,\sqrt{2}}{2\,Y\left(t,r\right)\left|\sin\left(\theta\right)\right|}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{1}{2}\,\sqrt{2}Y\left(t,r\right)\frac{\partial}{\partial r}Y\left(t,r\right) + \frac{\sqrt{2}}{2\,Y\left(t,r\right)^{3}\,h\left(t\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{1}{2}\,\sqrt{2}Y\left(t,r\right)\frac{\partial}{\partial r}Y\left(t,r\right) - \frac{\sqrt{2}}{2\,Y\left(t,r\right)^{3}h\left(t\right)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)}{4\,Y\left(t,r\right)\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)}{4\,Y\left(t,r\right)\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}Y\left(t,r\right)^{2}\frac{\partial^{2}}{\partial t\partial r}Y\left(t,r\right)}{4\,\frac{\partial}{\partial t}Y\left(t,r\right)} + \frac{1}{2}\,\sqrt{2}Y\left(t,r\right)\frac{\partial}{\partial r}Y\left(t,r\right) + \frac{\sqrt{2}}{2\,Y\left(t,r\right)^{3}h\left(t\right)}$$

$$\gamma = \frac{\sqrt{2}Y\left(t,r\right)^{2}\frac{\partial^{2}}{\partial t\partial r}Y\left(t,r\right)}{4\,\frac{\partial}{\partial t}Y\left(t,r\right)} + \frac{1}{2}\,\sqrt{2}Y\left(t,r\right)\frac{\partial}{\partial r}Y\left(t,r\right) - \frac{\sqrt{2}}{2\,Y\left(t,r\right)^{3}h\left(t\right)}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{Y\left(t,r\right)^{3}\frac{\partial^{2}}{\partial t\partial r}Y\left(t,r\right)\frac{\partial}{\partial r}Y\left(t,r\right)}{2\,\frac{\partial}{\partial t}Y\left(t,r\right)} - \frac{1}{2}\,Y\left(t,r\right)^{3}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(t,r\right) + \frac{\frac{\partial}{\partial t}h\left(t\right)}{2\,Y\left(t,r\right)^{5}h\left(t\right)^{3}\frac{\partial}{\partial t}Y\left(t,r\right)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{Y(t,r)^4 \frac{\partial^3}{\partial t (\partial r)^2} Y(t,r)}{4 \frac{\partial}{\partial t} Y(t,r)} + \frac{3\, Y(t,r)^3 \frac{\partial^2}{\partial t \partial r} Y(t,r) \frac{\partial}{\partial r} Y(t,r)}{2 \frac{\partial}{\partial t} Y(t,r)} + \frac{5}{4} Y(t,r)^2 \frac{\partial}{\partial r} Y(t,r)^2 + \frac{1}{2} Y(t,r)^3 \frac{\partial^2}{(\partial r)^2} Y(t,r) + \frac{1}{4\, Y(t,r)^2} - \frac{9}{4\, Y(t,r)^6 h(t)^2} - \frac{\frac{\partial}{\partial t} h(t)}{2\, Y(t,r)^5 h(t)^3 \frac{\partial}{\partial t} Y(t,r)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{Y(t,r)^3 \frac{\partial^2}{\partial t \partial r} Y(t,r) \frac{\partial}{\partial r} Y(t,r)}{2 \frac{\partial}{\partial t} Y(t,r)} - \frac{1}{2} Y(t,r)^3 \frac{\partial^2}{(\partial r)^2} Y(t,r) + \frac{\frac{\partial}{\partial t} h(t)}{2\, Y(t,r)^5 h(t)^3 \frac{\partial}{\partial t} Y(t,r)}$$

$$\Lambda = -\frac{Y(t,r)^4 \frac{\partial^3}{\partial t (\partial r)^2} Y(t,r)}{12 \frac{\partial}{\partial t} Y(t,r)} - \frac{2\, Y(t,r)^3 \frac{\partial^2}{\partial t \partial r} Y(t,r) \frac{\partial}{\partial r} Y(t,r)}{3 \frac{\partial}{\partial t} Y(t,r)} - \frac{5}{4} Y(t,r)^2 \frac{\partial}{\partial r} Y(t,r)^2 - \frac{1}{3} Y(t,r)^3 \frac{\partial^2}{(\partial r)^2} Y(t,r) + \frac{1}{12\, Y(t,r)^2} + \frac{1}{4\, Y(t,r)^6 h(t)^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{Y(t,r)^4 \frac{\partial^3}{\partial t (\partial r)^2} Y(t,r)}{6 \frac{\partial}{\partial t} Y(t,r)} + \frac{5\, Y(t,r)^3 \frac{\partial^2}{\partial t \partial r} Y(t,r) \frac{\partial}{\partial r} Y(t,r)}{6 \frac{\partial}{\partial t} Y(t,r)} + \frac{1}{2} Y(t,r)^2 \frac{\partial}{\partial r} Y(t,r)^2 + \frac{1}{6} Y(t,r)^3 \frac{\partial^2}{(\partial r)^2} Y(t,r) - \frac{1}{6\, Y(t,r)^2} - \frac{5}{2\, Y(t,r)^6 h(t)^2} - \frac{\frac{\partial}{\partial t} h(t)}{2\, Y(t,r)^5 h(t)^3 \frac{\partial}{\partial t} Y(t,r)}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"